

7.1.1 Magnetic Effects of Current Flow :

- In the electrostatics we find electric field lines originate on positive charge and terminate on negative charge.
- Such charges are absent in case of magnetic, due to which magnetic field lines can never have a beginning or an end. Magnetic field lines always form closed loops.

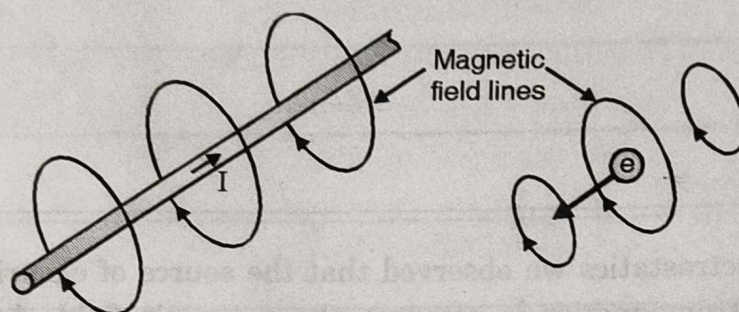
Magnetic field lines are always close in nature

- Without magnetic charges, magnetic fields can only arise indirectly.

- In fact, all magnetic fields are generated indirectly by moving electric charges.
- It is a fundamental fact of nature that moving electrons, as well as any other charges, produce a magnetic field in motion.

Any charge in motion produces magnetic field

- Then the question arises, how the current carrying filament produces magnetic field?
- The answer is, current is basically the collective movement of large number of electrons.
- A steady (DC) current through a wire produces a magnetic field that encircles the wire, as shown in Fig. 7.1.1(a).



(a) Field due to current

(b) field due to charge in motion

Fig. 7.1.1 : Production of magnetic field

- A single charge moving at constant velocity also produces a tubular magnetic field that encircles the charge, as shown in Fig. 7.1.1(b).
- However, the field of a single charge decays along the axis of propagation, with the maximum field occurring in the neighbourhood of the charge.
- Here note that when the velocity is constant then the acceleration is ($a = dv/dt$) zero. But if the charge has some acceleration it will not produce the magnetic field.
- The law that describes the field is called the Biot-Savart law, named after two French scientists who discovered it.
- Since one of the source of magnetic field is current, in the next article we discuss current and current distributions.

7.1.2 Current Distributions :

- The most common type of conductor used in practice is circular in cross section of radius 'a'. It carries a total current I as shown in Fig. 7.1.2(a).
- If the radius of the conductors is very small then it is known as a filamentary, Fig. 7.1.2(b).
- Fig. 7.1.2(b) shows a very small section of length $d\vec{l}$ of a filamentary conductor carrying a current I in $\vec{a}_l$ direction.
- Then a differential current element for a filamentary conductor is given by $I d\vec{l}$, where $d\vec{l} = d\vec{l} \vec{a}_l$.

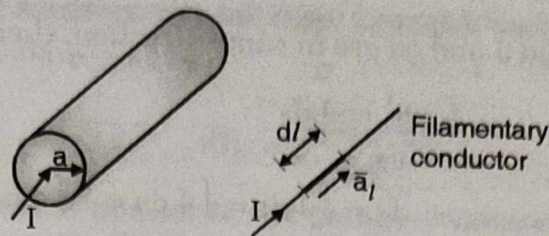
(a) Circular conductor current I (b) Filamentary conductor carrying current I

Fig. 7.1.2

Now we discuss three types of current densities :

- i) Uniform current density
- ii) Non uniform current density
- iii) Surface current density

i) Uniform current density :

Fig. 7.1.3(a) shows a rectangular slab of height h and width b . It carries a total current of I in $\bar{a}$ direction which is perpendicular to the cross sectional area (bh). The current density $\bar{J}$ is measured in A/m^2 , given by

$$\bar{J} = \frac{I}{bh} \bar{a} \quad A/m^2 \quad \dots(7.1.1)$$

and the total current if J is uniform is given by

$$I = J(bh) \quad \dots(7.1.2)$$

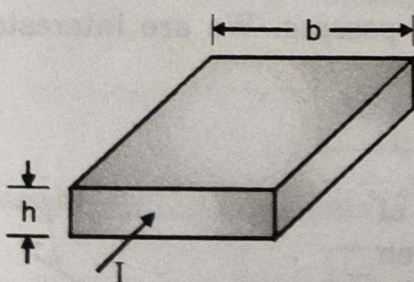
ii) Non uniform current density :

For a non uniform current density $\bar{J}$, when the direction of current is not perpendicular to cross section then integration of dot product $\bar{J} \cdot d\bar{s}$ is necessary.0 (Fig. 7.1.3(c))

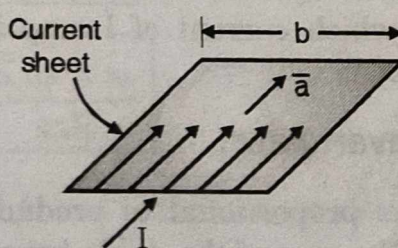
i.e.

$$I = \int_s \bar{J} \cdot d\bar{s}$$

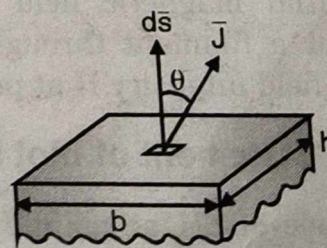
...(7.1.3)



(a) Conductor, rectangular in cross section



(b) Current sheet



(c) Non-uniform current density

Fig. 7.1.3

Consider now J is uniform and $\vec{J}$ and $d\vec{s}$ are in same direction, then

$$\vec{J} \cdot d\vec{s} = J ds$$

and

$$I = \int_S \vec{J} \cdot d\vec{s} = \int_S J ds = J \int_S ds = J (bh)$$

which is same as equation (7.1.2).

iii) Surface current density :

Fig. 7.1.3(b) shows a rectangular slab with height h very small. Then, current I will flow on a surface called as surface current and the surface current density measured in amperes per meter width is designated by $\vec{K}$. If we assume that width 'b' is measured perpendicular to the direction in which the current is flowing, then surface current density $\vec{K}$ is given by,

$$\vec{K} = \frac{I}{b} \vec{a} \text{ A/m} \quad \dots(7.1.4)$$

If the surface current density is uniform, the total current I in any width 'b' is,

$$I = Kb \quad \dots(7.1.5)$$

Again for a non uniform surface current density, integration is necessary, therefore,

$$I = \int K dl \quad \dots(7.1.6)$$

where, dl is a differential element of the path across which the current is flowing. Thus, the differential current element $I d\vec{l}$, where $d\vec{l}$ is in the direction of the current, may be expressed in terms of surface current density $\vec{K}$ or current density as

$$I d\vec{l} = \vec{K} ds = \vec{J} dv \quad \dots(7.1.7)$$

7.2 Biot-Savart Law :

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This law is also called as **Ampere's law** for the current element. It gives differential magnetic field intensity, $d\vec{H}$ due to differential current element $I d\vec{l}$. Consider a filament through which current of I amp is passing. We are interested in finding field intensity H at point P .

7.2.1 Statement of Biot Savart Law :

Magnitude of dH at point P is proportional to product of current, differential length dl , the sine of the angle between the filament and line connecting differential length to the point of interest P . It is inversely proportional to the square of the distance from the filament to point P . The constant of proportionality is $1/4 \pi$.

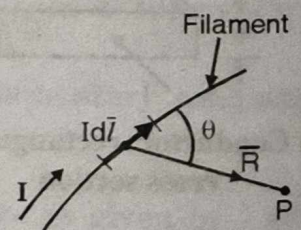


Fig. 7.2.1 : Biot-Savart law

7.2.2 Mathematical Expression for Biot Savart Law :

Mathematically,

$$dH = \frac{(Idl) \sin \theta}{4\pi R^2} \quad \dots(7.2.1)$$

The direction of dH is normal to the plane containing the differential element and the line drawn from the filament to the point P.

Of the two possible normals that normal is chosen which is in the direction of progress of a right handed screw turned from $d\vec{l}$ through the smaller angle to the line from the filament to P.

In vector notations,

$$d\vec{H} = \frac{Id\vec{l} \times \vec{a}_R}{4\pi R^2} \text{ (A/m)} \quad \dots(7.2.2)$$

where, $\times$ = this sign is a cross product.

$\vec{a}_R$ = unit vector from differential current element to point P.

$Id\vec{l}$ = differential current element.

R = distance of differential current element from point P.

- Differential current elements have no separate existence. All elements making up the complete current filament contribute to $\vec{H}$ and must be included. The summation leads to the integral form of the Biot-savart law as,

$$\vec{H} = \oint \frac{Id\vec{l} \times \vec{a}_R}{4\pi R^2} \quad \dots(7.2.3)$$

- The above integral is a closed line integral, which is must to ensure that all current elements are included (the contour may close at ∞)
- Alternate forms of the Biot-savart law in terms of distributed sources like $\vec{J}$ and $\vec{K}$ are obtained using Equation (7.1.7) as,

$$d\vec{H} = \frac{\vec{J} dv \times \vec{a}_R}{4\pi R^2} \quad \dots(7.2.4)$$

and

$$d\vec{H} = \frac{\vec{K} ds \times \vec{a}_R}{4\pi R^2} \quad \dots(7.2.5)$$

- Integrating Equations (7.2.4) and (7.2.5) we get total $\vec{H}$.

i.e.
$$\vec{H} = \int d\vec{H} \quad \dots(7.2.6)$$

- Note that, $d\vec{H}$ inside the integral is from Equations (7.2.2), (7.2.4) or (7.2.5) then the integral is single integral, tripple integral or double integral, which is very clear from $d\vec{l}$, dv and ds in the expressions for dH .

7.2.3 Solved Examples on Biot Savart Law :

Important Formulae

$$d\vec{H} = \frac{I d\vec{l} \times \vec{a}_R}{4\pi R^2} \text{ (A/m)}$$

$$d\vec{H} = \frac{\vec{K} ds \times \vec{a}_R}{4\pi R^2}$$

$$d\vec{H} = \frac{\vec{J} dv \times \vec{a}_R}{4\pi R^2}$$

$$\vec{H} = \int d\vec{H}$$

Example 7.2.1 : Estimate the incremental field $d\vec{H}_2$ at point ' P_2 ' caused by a source at ' P_1 ' of $I_1 d\vec{L}_1$:

- (i) $2\pi \vec{a}_z \mu\text{A} - \text{mt}$, given $P_1(4,0,0)$ and $P_2(0,3,0)$.
 (ii) $2\pi \vec{a}_z \mu\text{A} - \text{mt}$, given $P_1(4,-2,3)$ and $P_2(0,3,0)$.

Solution :

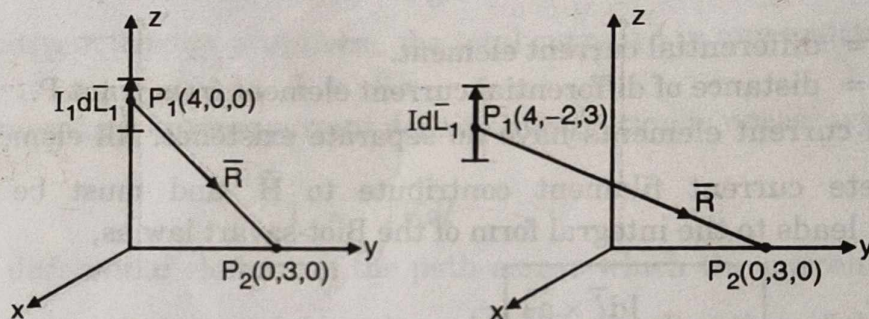


Fig. : Illustrating example 7.2.1, (a) part (i), (b) part (ii)

(i) We have,
$$d\vec{H}_2 = \frac{I d\vec{L} \times \vec{a}_R}{4\pi R^2}$$

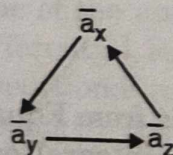
where,
$$\vec{R} = (0-4)\vec{a}_x + (3-0)\vec{a}_y + 0\vec{a}_z = -4\vec{a}_x + 3\vec{a}_y$$

 and
$$\vec{a}_R = \frac{-4\vec{a}_x + 3\vec{a}_y}{5}$$

Putting these in the expression for $d\vec{H}_2$ we get,

$$d\vec{H}_2 = \frac{2\pi \times 10^{-6} \vec{a}_z}{4\pi (5)^2} \times \left(\frac{-4\vec{a}_x + 3\vec{a}_y}{5} \right) = 4 \times 10^{-9} \vec{a}_z \times (-4\vec{a}_x + 3\vec{a}_y)$$

Using the triangle of cross product



we find :

$$\bar{a}_z \times \bar{a}_x = \bar{a}_y \text{ and } \bar{a}_z \times \bar{a}_y = -\bar{a}_x$$

$$\therefore d\bar{H}_2 = -12 \bar{a}_x - 16 \bar{a}_y \text{ (nA/m)}$$

...Ans.

(ii) In Fig. (b)

$$\bar{R} = (0-4) \bar{a}_z + (3+2) \bar{a}_y + (0-3) \bar{a}_x = -4 \bar{a}_x + 5 \bar{a}_y - 3 \bar{a}_z$$

$$\text{and } \bar{a}_R = \frac{-4\bar{a}_x + 5\bar{a}_y - 3\bar{a}_z}{5\sqrt{2}}$$

Then we get $d\bar{H}_2$ as

$$d\bar{H}_2 = \frac{2\pi \times 10^{-6} \bar{a}_z}{4\pi (5\sqrt{2})^2} \times \left(\frac{-4\bar{a}_x + 5\bar{a}_y - 3\bar{a}_z}{4\pi 5\sqrt{2}} \right) = 1.414 \bar{a}_z \times (-4\bar{a}_x + 5\bar{a}_y - 3\bar{a}_z)$$

Using the triangle of cross product we find : $\bar{a}_z \times \bar{a}_x = \bar{a}_y$; $\bar{a}_z \times \bar{a}_y = -\bar{a}_x$

and we know $\bar{a}_z \times \bar{a}_z = 0$

$$\therefore d\bar{H}_2 = -7.071 \bar{a}_x - 5.657 \bar{a}_y \text{ (nA/m)}$$

...Ans.

Example 7.2.2 : Given points A (1,2,4), B (-2,-1,3) and C (3,1,-2) let a differential current element with $I = 6A$ and $|dL| = 10^{-4}m$ be located at A. The direction of dL is from A to B. Find dH at C.

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Solution : The direction of dL is from A to B, then

$$I d\bar{l} = (6) (10^{-4}) (\bar{a}_{AB})$$

$$\text{Here } \bar{AB} = (-2-1) \bar{a}_x + (-1-2) \bar{a}_y + (3-4) \bar{a}_z = -3 \bar{a}_x - 3 \bar{a}_y - \bar{a}_z$$

Then the unit vector

$$\bar{a}_{AB} = \frac{\bar{AB}}{AB} = \frac{-3 \bar{a}_x - 3 \bar{a}_y - \bar{a}_z}{\sqrt{(-3)^2 + (-3)^2 + (-1)^2}} = \frac{-3 \bar{a}_x - 3 \bar{a}_y - \bar{a}_z}{\sqrt{19}}$$

$$\begin{aligned} \therefore I d\bar{l} &= (6) (10^{-4}) \left(\frac{-3 \bar{a}_x - 3 \bar{a}_y - \bar{a}_z}{\sqrt{19}} \right) \\ &= -0.413 \bar{a}_x - 0.413 \bar{a}_y - 0.138 \bar{a}_z \text{ (mA}\cdot\text{m)}. \end{aligned}$$

The Biot-Savart law is

$$d\bar{H} = \frac{I d\bar{l} \times \bar{a}_{AC}}{4\pi(AC)^2}$$

$$\text{Now } \bar{AC} = (3-1) \bar{a}_x + (1-2) \bar{a}_y + (-2-4) \bar{a}_z = 2 \bar{a}_x - \bar{a}_y - 6 \bar{a}_z$$

$$\bar{a}_{AC} = \frac{\bar{AC}}{AC} = \frac{2 \bar{a}_x - \bar{a}_y - 6 \bar{a}_z}{\sqrt{2^2 + (-1)^2 + (-6)^2}} = \frac{2 \bar{a}_x - \bar{a}_y - 6 \bar{a}_z}{\sqrt{41}}$$

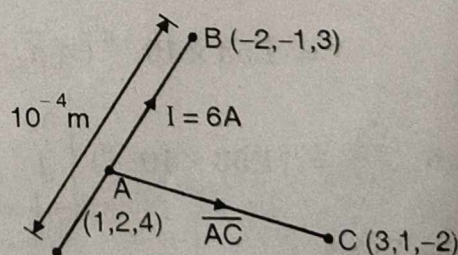


Fig. Illustrating Ex. 7.2.2

(c) **Direction of $\bar{a}_l$:**

To have $d\bar{H} = 0$ at P, the numerator of $d\bar{H}$ of Equation (A) must be zero. The cross product of two vectors is zero if vectors are collinear with each other. That is,

$$\bar{a}_l = \pm \bar{a}_R = \frac{\pm (-\bar{a}_x + \bar{a}_y - \bar{a}_z)}{\sqrt{3}}$$

The Biot-Savart law is only applicable to differential current filament. But in practice the filament may be finite length or circular etc. The standard type of current filament is infinite long straight filament, while finding H due to these using Biot savart law, the filament can be broken into number of differential elements and superposition can be used to find total H. Magnetic field intensities due to some standard filaments are derived in the next few sections.

Example 7.2.4 : Using Biot-Savart law, find the vector magnetic field intensity in cartesian coordinates at $P_2 (1.5, 2, 3)$ caused by a current filament of 24 A in the $\bar{a}_z$ direction on the z-axis and extending from : (a) $z = 0$ to $z = 6$; (b) $z = 6$ to $z = \infty$; (c) $z = -\infty$ to $z = \infty$.

Solution :

Let the differential current filament on z-axis is at $(0, 0, z)$. The vector joining this point and point P_2 is

$$\bar{R} = 1.5 \bar{a}_x + 2 \bar{a}_y + (3 - z) \bar{a}_z$$

$$\bar{a}_R = \frac{1.5 \bar{a}_x + 2 \bar{a}_y + (3 - z) \bar{a}_z}{\sqrt{1.5^2 + 2^2 + (3 - z)^2}} = \frac{1.5 \bar{a}_x + 2 \bar{a}_y + (3 - z) \bar{a}_z}{\sqrt{(6.25) + (3 - z)^2}}$$

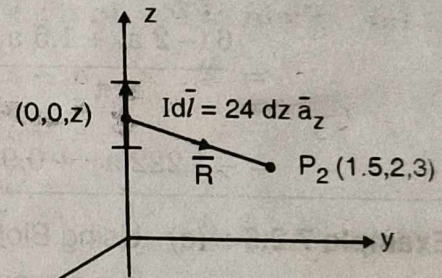


Fig. : Illustrating Ex. 7.2.4

Using Biot-Savart law,

$$\begin{aligned} \therefore d\bar{H}_2 &= \frac{I d\bar{l} \times \bar{a}_R}{4 \pi R^2} \\ &= \frac{24 dz \bar{a}_z}{4 \pi (6.25 + (3 - z)^2)} \times \frac{1.5 \bar{a}_x + 2 \bar{a}_y + (3 - z) \bar{a}_z}{\sqrt{(6.25) + (3 - z)^2}} = \frac{6 (-2 \bar{a}_x + 1.5 \bar{a}_y) dz}{\pi [6.25 + (3 - z)^2]^{3/2}} \end{aligned}$$

(a) **For $z = 0$ to $z = 6$:**

The total magnetic field intensity is obtained by integrating $d\bar{H}_2$.

$$\bar{H}_2 = \int_0^6 d\bar{H}_2 = \frac{6 (-2 \bar{a}_x + 1.5 \bar{a}_y)}{\pi} \int_0^6 \frac{dz}{[6.25 + (3 - z)^2]^{3/2}}$$

$$= \frac{6(-2\bar{a}_x + 1.5\bar{a}_y)}{\pi} \times \left[-\frac{1}{6.25} \times \frac{(3-z)}{\sqrt{6.25 + (3-z)^2}} \right]_0^6 = 0.469(-2\bar{a}_x + 1.5\bar{a}_y)$$

$$= -0.939\bar{a}_x + 0.704\bar{a}_y \text{ (A/m)} \quad \dots\text{Ans.}$$

(b) For $z = 6$ to $z = \infty$:

$$\bar{H}_2 = \int_6^{\infty} d\bar{H}_2 = \frac{6(-2\bar{a}_x + 1.5\bar{a}_y)}{\pi} \int_6^{\infty} \frac{dz}{[6.25 + (3-z)^2]^{3/2}}$$

$$= \frac{6(-2\bar{a}_x + 1.5\bar{a}_y)}{\pi} \left[-\frac{1}{6.25} \times \frac{(3-z)}{\sqrt{6.25 + (3-z)^2}} \right]_6^{\infty}$$

$$= -0.1417\bar{a}_x + 0.1062\bar{a}_y \text{ (A/m)} \quad \dots\text{Ans.}$$

(c) For $z = -\infty$ to ∞ :

$$\bar{H}_2 = \int_{-\infty}^{\infty} d\bar{H}_2 = \frac{6(-2\bar{a}_x + 1.5\bar{a}_y)}{\pi} \int_{-\infty}^{\infty} \frac{dz}{[6.25 + (3-z)^2]^{3/2}}$$

$$= \frac{6(-2\bar{a}_x + 1.5\bar{a}_y)}{\pi} \left[-\frac{1}{6.25} \times \frac{(3-z)}{\sqrt{6.25 + (3-z)^2}} \right]_{-\infty}^{\infty}$$

$$= -1.222\bar{a}_x + 0.917\bar{a}_y \text{ (A/m)} \quad \dots\text{Ans.}$$

- Example 7.2.5 :** (a) Using Biot-Savart law find $\bar{H}$ in cartesian components at P (2,3,4) if there is a current filament on the z-axis carrying 8 mA in the $\bar{a}_z$ direction.
 (b) Repeat if the filament is located at $x = -1, y = 2$.
 (c) Find $\bar{H}$ if both filaments are present.

Solution :

(a) $\bar{H}$ due to filament on the z-axis :

Let the differential current filament on z axis is at $(0, 0, z)$. The vector and unit vector from $(0, 0, z)$ towards point P are (refer Fig. Ex. 7.2.5(a))

$$\bar{R} = 2\bar{a}_x + 3\bar{a}_y + (4-z)\bar{a}_z$$

$$\bar{a}_R = \frac{2\bar{a}_x + 3\bar{a}_y + (4-z)\bar{a}_z}{\sqrt{2^2 + 3^2 + (4-z)^2}}$$

$$= \frac{2\bar{a}_x + 3\bar{a}_y + (4-z)\bar{a}_z}{\sqrt{(4-z)^2 + 13}}$$

and $R = \sqrt{(4-z)^2 + 13}$

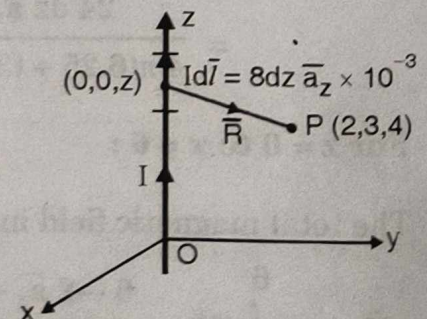


Fig. : Illustrating Ex. 7.2.5(a)